Predicting Coronary Heart Disease from Survey Data with Linear Models

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Code: https://github.com/Candicci/ML_Project_HeartAttack

***Abstract*—We predict whether a person has been diagnosed with coronary heart disease (MICHD) from 321 health and lifestyle features of the BRFSS telephone survey. Early detection of cardiovascular disease matters, but the task is hard: only 8.8% of the respondents are positive and about 45% of the feature values are missing. We implement six linear methods from scratch (least squares, ridge, linear regression with GD and SGD, and logistic regression with and without regularization) and compare them in one shared pipeline. With the default decision cutoff all of them reach an F1 score of at most 0.18. Tuning the cutoff on a validation set raises it to 0.39–0.41 for every method, which matters much more than the choice of model. Our logistic regression scores F1 = 0.416 on the competition test set.**

I. Introduction

The dataset contains 328,135 labelled respondents (321 features) and 109,379 unlabelled ones for the competition. The label is $+1$ if the respondent reported MICHD (heart attack or angina) and -1 otherwise. Because positives are rare, a classifier that always predicts -1 already reaches 91% accuracy, so we use the F1 score of the positive class as the main metric.

Our contributions are: (i) correct implementations of the six required methods, all trained and tuned with the same preprocessing and protocol; (ii) an ablation showing that the decision cutoff is the most important choice for this imbalanced problem; and (iii) evidence that all linear models perform alike once the cutoff is tuned, and that none of them overfits.

II. Models and Methods

A. Data and preprocessing

We split the labelled data at random (seed 42) into train / validation / test sets of 80/10/10% (262,508 / 32,813 / 32,814 rows). The positive rate is 8.8–8.9% in each split. All statistics are computed on the train split only. We drop the 6 columns that are constant or entirely missing, standardize each remaining column to zero mean and unit variance, replace missing values by 0 (i.e. the train mean), and add a bias column. This gives $D = 316$ inputs. The same transformation is applied to the validation, test and competition sets. Standardization is essential: some raw columns reach $\sim 10^9$, which makes the sigmoid overflow.

B. Models

Let $\mathbf{x}_n \in \mathbb{R}^D$ be the preprocessed features and N the number of training rows. The linear-regression methods use labels $y_n \in \{-1, 1\}$ and the MSE loss; the logistic methods use $y_n \in \{0, 1\}$ and the negative log-likelihood:

$$L_{\text{MSE}}(\mathbf{w}) = \tfrac{1}{2N} \textstyle\sum_n (y_n - \mathbf{x}_n^\top \mathbf{w})^2,$$
$$L_{\text{NLL}}(\mathbf{w}) = \tfrac{1}{N} \textstyle\sum_n \left[\log(1 + e^{\mathbf{x}_n^\top \mathbf{w}}) - y_n \mathbf{x}_n^\top \mathbf{w} \right].$$

Least squares solves the normal equations $X^\top X \mathbf{w} = X^\top \mathbf{y}$ and *ridge regression* solves $(X^\top X + 2N\lambda I)\mathbf{w} = X^\top \mathbf{y}$. *GD* and *SGD* (one random sample per step) minimize L_{MSE}. *Logistic regression* runs gradient descent on L_{NLL}, and the *regularized* version adds $\lambda \|\mathbf{w}\|^2$ to the objective. We compute $\log(1 + e^z)$ with `np.logaddexp` so that it cannot overflow. All methods predict $\hat{y} = +1$ if $\mathbf{x}^\top \mathbf{w} \geq t$ and -1 otherwise, with cutoff t.

C. Training and evaluation protocol

All iterative methods start from $\mathbf{w}_0 = \mathbf{0}$. GD and both logistic methods run 200 full-gradient iterations. SGD runs 10 epochs of N single-sample steps. We sweep the step size γ on log grids: $\{10^{-3}, \ldots, 10^{-1}\}$ for GD, $\{10^{-4}, \ldots, 3{\cdot}10^{-3}\}$ for SGD and $\{10^{-2}, \ldots, 1\}$ for logistic regression. For ridge we sweep 17 values $\lambda \in [10^{-8}, 1]$, and for regularized logistic regression $\lambda \in \{10^{-5}, \ldots, 10^{-1}\}$, using the best logistic γ. Each hyperparameter is chosen by the lowest *validation* loss. We obtain loss curves at no extra cost by running the methods in chunks of iterations and restarting each chunk from the previous $\mathbf{w}$; this gives exactly the same weights as a single run.

The cutoff t is then tuned per method on the validation set: we try 200 quantiles of the validation scores $\mathbf{x}^\top \mathbf{w}$ and keep the one with the highest F1. The final method is the one with the best validation F1. The test split is used only to report results. The notebook `run.ipynb` reproduces all numbers and the submitted predictions.

III. Results

Table I summarizes the test results. **The cutoff is the dominant factor.** With $t = 0$ the linear-regression models almost never predict $+1$ (F1 ≤ 0.02), and the logistic models only reach F1 = 0.18. Tuning t raises F1 by 0.23 (logistic) to 0.40 (GD). All tuned cutoffs are negative (≈ -0.57 for MSE scores, -1.48 to -1.54 for logistic scores), meaning

Method	Hyperparameters	Acc.	F1 ($t = 0$)	F1 (tuned t)
GD	$\gamma = 0.03$	0.860	0.004	0.400
SGD	$\gamma = 10^{-4}$	0.865	0.007	0.386
Least squares	–	0.864	0.017	0.411
Ridge	$\lambda = 3 \cdot 10^{-6}$	0.862	0.016	0.412
Logistic	$\gamma = 1$	0.858	0.183	**0.413**
Reg. logistic	$\gamma = 1$, $\lambda = 10^{-5}$	0.851	0.183	0.412

Table I
RESULTS ON OUR LOCAL TEST SPLIT (32,814 ROWS). ACCURACY IS AT THE TUNED CUTOFF. LOGISTIC REGRESSION (CUTOFF $t = -1.48$) WAS SELECTED ON VALIDATION F1 (0.417) AND SCORED F1 = 0.416, ACCURACY = 0.860 ON THE COMPETITION TEST SET.

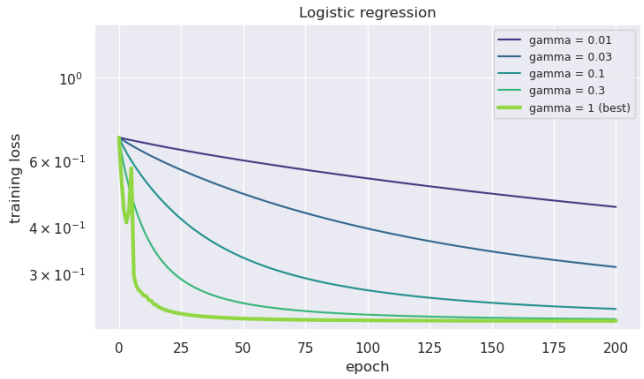


Figure 1. Training loss (L_{NLL}, log scale) of logistic regression for each step size γ; one epoch is one full-gradient iteration.

that many more people must be flagged than with the default cutoff. The price is a small loss in accuracy: 0.85–0.87, compared with 0.91 for always predicting -1. After tuning, the six methods lie within 0.03 F1 of each other; logistic regression wins narrowly and generalizes to the competition test set (0.416 against 0.413 locally).

Step size. Figure 1 shows that small step sizes converge slowly, while $\gamma = 1$ converges fastest after a short initial instability. For the MSE methods, too large a step size diverges: GD with $\gamma = 0.1$ and SGD with $\gamma \geq 3 \cdot 10^{-4}$ (the training loss reaches 10^{139} and then `inf`). SGD needs a much smaller γ because each step is based on a single noisy sample.

Regularization has almost no effect. Ridge validation loss is flat for $\lambda \leq 10^{-3}$, and the best regularized logistic model uses the smallest λ in the grid. With $N \approx 262{,}000 \gg D = 316$ there is little variance to reduce, and train and test F1 differ by at most 0.007 for every model.

Least squares and conditioning. The design matrix has rank 310 of 316 (linearly dependent columns), so $X^\top X$ is singular (cond $\approx 9 \cdot 10^{18}$). On the full data the solver still returns usable weights, but in an early experiment on a 5,250-row subsample the same solver gave a validation MSE of $4 \cdot 10^4$. Ridge with any $\lambda > 0$ removes this problem.

IV. DISCUSSION

Train and test performance are almost identical, so our models *underfit* rather than overfit: a linear decision rule on the raw survey codes is the limiting factor, not the amount of data or the regularization. This is consistent with the six very different training methods converging to the same F1. Two weaknesses of our setup point to the next steps. First, many BRFSS features are categorical codes (e.g. 7/9 for "don't know/refused", 88 for "none") that we treat as numbers; one-hot encoding or recoding these values, together with missing-value indicators, should give the linear models more useful features. Second, we use a single train/validation/test split. The splits are large (about 33,000 rows each) and validation and test F1 agree, but k-fold cross-validation would give the standard deviation of our estimates. Finally, the selected $\gamma = 1$ lies at the edge of its grid, so a wider sweep or more iterations could still give a small gain.

V. SUMMARY

We implemented and compared six linear methods for predicting coronary heart disease on a large, imbalanced and incomplete survey dataset. With careful standardization all methods train stably, and tuning the decision cutoff on a validation set is by far the most effective change, raising F1 from at most 0.18 to about 0.41. Our selected logistic regression model reaches F1 = 0.416 and accuracy = 0.860 on the competition test set.